Below is a **master cloud terminology list** used across **AWS, Microsoft Azure, and Google Cloud (GCP)** (covering **core + advanced terms**), with **real-world use cases + examples**, in a **table format**.

✅ Note: Cloud has 1000s of terms. So I'm listing the **most-used + most-important industry terminology** that covers **almost all real software/cloud architectures**.

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1) Cloud Fundamentals (Common across AWS / Azure / GCP)

Terminology	Meaning	Real-time Use Case	Example in Software/App
Cloud Computing	On-demand computing resources via internet	Run apps without owning servers	Hosting an e-commerce website
IaaS	Infrastructure as a Service (VMs, networks)	Need full control over OS	Running legacy Java app on VM
PaaS	Platform as a Service (runtime managed)	Developer focus, infra managed	Deploy Spring Boot app without servers
SaaS	Software as a Service	Use software directly	Gmail, Office365
Serverless	Run code without managing servers	Event-driven workloads	Image upload triggers processing
Pay-as-you-go	Billing based on usage	Cost optimized scaling	Pay per request in serverless APIs
Multi-Tenancy	Sharing cloud infra among customers	Efficient SaaS architecture	HR SaaS app used by many companies
Cloud Region	Geographic location of data centers	Low latency, compliance	India users served from Mumbai region
Availability Zone (AZ)	Isolated datacenters inside a region	High availability setup	DB in multi-AZ deployment
Edge Location	CDN/edge nodes near users	Reduce latency	Serving videos to global users
Elasticity	Auto scale up/down	Handle traffic spikes	Sale/offer traffic surge scaling
Scalability	Increase capacity	Growth-ready architecture	App scaling from 1k → 1M users

High Availability (HA)	System stays online during failure	Business continuity	Multi-zone load balancing
Fault Tolerance	Continue even when components fail	Critical systems	Banking payment system resiliency
Disaster Recovery (DR)	Recover from big failures	DR planning	Backup region deployment
RTO	Recovery Time Objective	Define max downtime	"App should recover in 15 mins"
RPO	Recovery Point Objective	Define max data loss allowed	"Max 5 mins data loss acceptable"
SLA	Service guarantee	Vendor reliability	99.9% uptime SLA
Shared Responsibility Model	Cloud provider vs customer duties	Security clarity	AWS manages infra; you manage app data

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2) Compute Terminology (VMs, Containers, Serverless)

Terminology	**Meaning**	**Use Case**	**Example**
VM / Virtual Machine	OS-level compute instance	Full control hosting	Java app on Ubuntu VM
Instance	Cloud VM object	Basic compute unit	EC2 VM instance
Auto Scaling	Increase/decrease instances	Handle variable load	10 servers at peak, 2 at night
Load Balancer	Distributes traffic	HA + scaling	API traffic balanced across servers
Container	Lightweight packaged app unit	Microservices deployment	Dockerized Spring Boot service
Container Registry	Store container images	CI/CD image storage	Store backend:v2.3
Kubernetes	Container orchestration	Microservices management	30 microservices deployment
Node Pool / Node Group	Group of Kubernetes nodes	Worker nodes scaling	Add nodes for new services

Pod	K8s smallest unit	Deploy workload	Spring Boot + sidecar logging
Helm	K8s package manager	Deploy complex apps	Install Kafka/Prometheus
FaaS / Function as a Service	Event-based code execution	Event automation	New file → run function
Cold Start	Serverless startup latency	Performance optimization	Function may take 1s first time
Runtime	Execution environment	Code execution	Java 21 runtime function
Managed Compute	Cloud manages underlying compute	Simple app hosting	App service deployment

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3) Storage Terminology (Object, Block, File)

Terminology	**Meaning**	**Use Case**	**Example**
Object Storage	Store files as objects	Media storage	Images/videos in cloud storage
Bucket / Container	Object storage grouping	Organize files	user-profile-images/ bucket
Block Storage	Disk attached to VM	DB/low latency apps	SSD disk for MySQL server
File Storage	Shared filesystem	Shared app content	Multiple servers share uploads
Snapshot	Disk backup copy	Restore quickly	VM disk snapshot before patch
Lifecycle Policy	Auto move/delete objects	Cost saving	Move logs to archive after 30 days
Versioning	Store old versions of files	Backup protection	Prevent accidental deletion
Archive Storage	Cheapest, slow retrieval	Long-term storage	Store compliance logs 7 years

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4) Networking Terminology (Core cloud network terms)

Terminology	Meaning	Use Case	Example
VPC / VNet	Private isolated network	Secure network design	App in private subnet
Subnet	Network segment	Security separation	Public subnet for LB only
Public IP	Internet-facing IP	External access	Public API endpoint
Private IP	Internal-only IP	Secure traffic	DB in private network
Route Table	Routing rules	Network flow control	Direct traffic to internet gateway
Internet Gateway	Connect VPC to internet	Public access	LB internet access
NAT Gateway	Private subnet outbound internet	Security	Private servers update packages
Firewall Rules	Allow/deny network traffic	Security control	Block all except port 443
Security Group / NSG	VM-level firewall	Network security	Allow SSH only from office IP
ACL / NACL	Subnet-level firewall	Network hardening	Block unknown CIDR ranges
DNS	Domain name resolution	Service discovery	api.myapp.com
CDN	Content delivery network	Faster global delivery	Serve static UI globally
VPN	Private connection	Hybrid connectivity	Office → cloud private network
Peering	Connect two private networks	Multi-VPC apps	Connect app VPC to analytics VPC
Private Endpoint	Access service privately	Secure service access	Access storage without internet
Service Mesh	Microservices networking	Observability + security	Istio mutual TLS between services

5) Database Terminology (SQL, NoSQL, Cache)

Terminology	Meaning	Use Case	Example
RDBMS	SQL database	Transactions	Banking account system
NoSQL	Document/Key-value DB	Flexible schema	Product catalog
Read Replica	Read-only copy	Performance scaling	Analytics queries run on replica
Multi-Master	Multiple writable nodes	Global write capability	Global distributed app
Sharding	Split DB across servers	Massive scale	1B users distributed DB
Partitioning	Split data logically	Performance	Monthly partitions for logs
OLTP	Transaction system	High frequency writes	Orders table
OLAP	Analytics workload	Reporting	BI dashboard queries
Data Warehouse	Analytics DB	BI reports	Weekly business insights
Cache	Memory fast access	Improve speed	Redis session storage
TTL	Expire cached data	Automatic cleanup	OTP expires in 2 mins
Index	Faster search	Optimize queries	Index on email column

6) Identity & Security Terminology

Terminology	Meaning	Use Case	Example
IAM	Identity & Access	Permissions	Dev can deploy but

	Management		not delete DB
Principle of Least Privilege	Minimal access	Security best practice	Allow only read permissions
Role	Temporary permissions	Secure access	App uses IAM role to access storage
Policy	Access rules	Fine control	Allow only bucket access
MFA	Multi-factor auth	Stop account takeover	MFA for admin accounts
Encryption at Rest	Encrypt stored data	Compliance	Encrypt DB disks
Encryption in Transit	Encrypt network data	Secure APIs	HTTPS TLS 1.2+
Key Management (KMS)	Manage encryption keys	Centralized encryption	Rotate keys every 90 days
Secrets Manager	Store credentials securely	Avoid hardcoding	DB password stored securely
Zero Trust	Never trust by default	Modern security model	Service-to-service auth always
WAF	Web App Firewall	Block attacks	Protect against SQL injection
DDoS Protection	Block traffic floods	Protect uptime	Shielding app during attack
SIEM	Security monitoring	Threat detection	Logs sent to SIEM for alerts

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7) Observability / Monitoring / Logging

Terminology	**Meaning**	**Use Case**	**Example**
Metrics	Numerical monitoring data	CPU/memory tracking	CPU > 80% alert
Logs	Text event records	Debugging	API failure logs
Traces	Request journey tracking	Microservices debugging	Trace slow requests

APM	Performance monitoring	detect bottlenecks	Find slow DB query
Alerting	Notification on issues	DevOps	PagerDuty alerts
SLO	Service objective	Reliability	99.95% monthly availability
SLA	Legal uptime guarantee	Business contract	99.9% uptime promised
SLIs	Measurements for SLO	Reliability metrics	95th percentile latency

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8) DevOps & CI/CD Terminology

Terminology	**Meaning**	**Use Case**	**Example**
CI/CD	Automated build/deploy	Fast releases	Deploy after Git push
Pipeline	Steps for automation	Controlled deployment	build → test → deploy
Artifact	Build output	Release packaging	JAR file created
Blue-Green Deploy	2 environments switch	Zero downtime updates	V1 runs, V2 ready then switch
Canary Deploy	Gradual rollout	Reduce risk	10% traffic to new version
Rollback	Restore old version	Quick recovery	Revert buggy release
IaC	Infrastructure as Code	Repeatable infra	Terraform deploy VPC+K8s
GitOps	Infra managed by Git	Auditable deployments	Kubernetes manifests in repo

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9) Messaging / Streaming Terminology

Terminology	Meaning	Use Case	Example
Message Queue	Async communication	Decouple services	Orders service → Payment queue
Pub/Sub	Broadcast messages	Event-driven apps	Notify many services on event
Event-Driven Architecture	Events trigger actions	Real-time automation	User signup triggers email
Consumer	Reads messages	Processing	Worker reads queue jobs
Producer	Sends messages	Async tasks	App produces order events
Topic	Message category	Organize events	order.created topic
Partition	Split topic for scale	High throughput	Kafka topic partitioning
Offset	Kafka message position	Replay	Start from offset 0 for reprocessing

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10) AWS vs Azure vs GCP Key Terminology Mapping (Most Important)

Concept	AWS Term	Azure Term	GCP Term
Cloud VM	EC2	Virtual Machines	Compute Engine
Auto Scaling	Auto Scaling Group	VM Scale Sets	Managed Instance Groups
Load Balancer	ELB/ALB/NLB	Azure Load Balancer / App Gateway	Cloud Load Balancing
Object Storage	S3	Blob Storage	Cloud Storage
Block Storage	EBS	Managed Disks	Persistent Disk
File Storage	EFS	Azure Files	Filestore
DNS	Route 53	Azure DNS	Cloud DNS

CDN	CloudFront	Azure CDN	Cloud CDN
Kubernetes	EKS	AKS	GKE
Container Registry	ECR	ACR	Artifact Registry
Serverless Functions	Lambda	Azure Functions	Cloud Functions
SQL DB (Managed)	RDS	Azure SQL Database	Cloud SQL
NoSQL DB	DynamoDB	Cosmos DB	Firestore/Bigtable
Data Warehouse	Redshift	Synapse Analytics	BigQuery
IAM	IAM	Azure AD / Entra ID	Cloud IAM
Monitoring	CloudWatch	Azure Monitor	Cloud Monitoring
Logging	CloudWatch Logs	Log Analytics	Cloud Logging
Secret Manager	Secrets Manager	Key Vault	Secret Manager
Messaging	SQS/SNS	Service Bus/Event Grid	Pub/Sub

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11) Real-time Cloud Architecture Examples (Practical)

Application Type	**Cloud Terms Used**	**Real-world Example**
E-commerce App	VM/Containers, LB, AutoScaling, DB, Cache, CDN	Amazon-like store
Microservices	Kubernetes, Service Mesh, Observability, CI/CD	Food delivery app
Video Platform	Object Storage, CDN, Streaming, Event-driven	YouTube-like video app
Banking System	HA, DR, Encryption, IAM, Audit Logs	UPI/Netbanking
IoT System	PubSub/Queue, Serverless, Data lake	Smart meter readings
Chat App	WebSockets, PubSub, Cache, NoSQL	WhatsApp-like chat
ML/AI App	Storage, GPU VMs, Pipelines, Feature Store	Face detection pipeline

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✅ If you want, I can also provide:

1. **A 500+ cloud terminology list** (AWS+Azure+GCP combined) in **Excel/CSV format**
2. **Interview-focused terminology with scenarios**
3. **A full cloud architecture diagram (PNG)**: Microservices + CI/CD + HA + DR + Observability